High Speed Camera — Work Instruction

Needle-free injector test rig — capture and measurement

Application	`HighSpeedCamera.exe`
Document version	1.0 — 2026-09-17
Applies to	Spray angle, piston speed, gel injection, monitoring

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1. What this app does

A high-speed camera watches a needle-free injector fire. The app records a short burst of images, finds the few milliseconds that actually contain the injection, and measures it.

There are **four jobs** it can do. Each is one button, and each runs its whole sequence automatically once you press it.

Job	Button	What you get
Spray angle	**Capture**	Angle of the jet vs the nozzle axis, per frame + a mean
Piston speed	**Capture Piston**	Distance/speed vs time for the piston stroke, plus jet speed and pressure
Gel injection	**Capture Gel Injection**	The injection images and a slow-motion video (no angle measurement)
Repeatability	**Start Monitoring**	Many injections in a row, each measured automatically

> **The camera is black and white.** There is no colour mode. Anything you need to see must differ in **brightness**, not in hue. A dye that is a different colour but the same lightness is invisible to this camera.

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2. Before you start

2.1 Every session

1. Connect the camera and start the app (`HighSpeedCamera.exe`).
2. The **Live Preview** starts automatically. Check the image is in focus and the injector is in frame.
3. Check the nozzle is positioned as it was when the reference was set (see 2.2) — if you moved the camera or the lens, redo section 2.2.

2.2 Calibration and reference — when they must be done

The camera cannot be used for measurement until both of these are done. They are stored and reused for every later run, so you only repeat them when something in the table below changes.

When must I redo them?

What changed	Calibrate Scale	Set Reference
Nothing — same rig, next shift	no	no
Camera moved (position, angle, height)	**YES**	**YES**
Lens changed or refocused	**YES**	**YES**
Working distance changed (camera↔nozzle)	**YES**	**YES**
Capture ROI changed (e.g. 320x240 → 320x120)	no	**YES** — see the warning below
Nozzle removed and refitted	no	**YES**
New PC, same camera and optics	no (copy `settings\`)	no (copy `settings\`)

> **Anything that changes the optics changes millimetres per pixel.** Moving the camera, swapping or refocusing the lens, or changing the distance to the nozzle all invalidate the calibration. Measuring with a stale calibration does not fail — it silently reports **confidently wrong millimetres**. When in doubt, recalibrate: it takes under a minute.

Do them in this order

Calibration **must** come first. The reference includes a target rectangle sized in millimetres, so it cannot be built without a scale. If you set the reference with no calibration stored, you get the lines but **no rectangle and no OK/NG verdict**.

Step 1 — Calibrate the scale (millimetres per pixel)

1. Set the **Capture ROI** you intend to use for the test.
2. Place the reference block in view.
3. Type its known size into **Cal. distance (mm)** — default `5`.
4. Press **Auto Calibrate**. - If it fails or the result looks wrong, press **Calibrate Scale** instead and drag a box around each of the two edges by hand, clicking *Detect Edge* for each.
5. **Check the saved** `*_scale.bmp`: the two green lines must sit exactly on the block's edges. This is the only check that catches a bad calibration.

> **Auto Calibrate always saves a new calibration**, taking the distance you typed as correct. **Measure Block** is the opposite — it measures an unknown block using the stored calibration and changes nothing. Calibrate on the known reference; measure everything else.

Step 2 — Set the nozzle reference (spray angle tests)

This defines the spray axis that the angle is measured against.

1. **Set the Capture ROI you will actually capture at** — see the warning below.
2. **The nozzle must be on the RIGHT-hand side of the frame.** This is fixed in the software and cannot be changed from the screen. A nozzle on the left will produce wrong angles.
3. Remove the jet from view. The nozzle must be **dry and still** — no mist, no droplets.

4. Press **Set Reference**. A still is captured and the dialog opens.
5. **Drag along the nozzle tip edge** to draw the red line. Auto-detection is unreliable on the reflective nozzle, so you place this line yourself — it is the single most important input to the spray angle.
6. The app builds the rest automatically: - the **blue centre line** through the tip midpoint, perpendicular to your red line — this is the spray axis - a **red target rectangle** on that axis, used for the OK/NG verdict
7. **Check the saved overlay image before closing.** If the red line is not on the tip edge, every angle from now on is measured against the wrong axis.

> **The reference only applies to frames of the same size it was set at.** If you set the reference at `320x240` and then capture at `320x120`, the app **silently ignores it**: the angle falls back to edge-to-edge instead of deviation from the spray axis, and the `in_rect` column comes out empty. Nothing warns you. **Set the reference at the ROI you will capture at**, and redo it whenever you change the ROI.
>
> **Changing the ROI does not spoil the calibration.** Millimetres per pixel is a property of the lens, not of how many sensor rows you read. You may see a note that the calibration was made on a different frame size — that is telling you the magnification is assumed unchanged, which is correct for a simple ROI crop. Only redo the calibration if the **optics** changed.

Step 3 — Confirm it took

Run one capture and open the measure folder. You should see:

- green edge lines sitting on the visible jet edges
- an `in_rect` column in the CSV reading `OK` or `NG` (not blank)
- results in **millimetres**, not pixels

If any of those is missing, go back to the matching step above.

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3. The screen

The window has the live preview on the right, and on the left, from top to bottom: **Capture Settings**, then three button groups.

3.1 Capture Settings

Field	Default	Notes
Capture time (s)	`4.0`	How long the burst records
Camera index	`1`	Leave at 1 unless you have more than one camera
Exposure (µs)	`500`	See the warning below
Capture ROI	`320x240`	`320x240` → **1048 fps**; `320x120` → **1798 fps**
Run label (optional)	empty	Free text, e.g. `IN39-30U` — added to every folder and file name
Output base folder	app folder	Where `captures`, `injection images` and `measure` are written

> **Exposure does not change the frame rate.** The rate is set by how many sensor **rows** are read, not by exposure time. 100 µs, 200 µs and 500 µs all give the same fps at a given ROI. So keep the exposure **high** for contrast, and shorten the **ROI** if you need more speed.
>
> **Choosing the ROI.** `320x120` is twice as fast and is the right choice for the piston stroke and for anything with a fast opening transient. `320x240` shows more of the scene. Changing the ROI restarts the preview. **Cropping does not change the calibration** — millimetres per pixel is a property of the lens, not of how many rows you read.

3.2 Initial Setup group

`Calibrate Scale` · `Auto Calibrate` · `Measure Block` · `Measure Angle Block` · `Set Reference`

Used in section 2.2. **Measure Block** is the inverse of calibration: it measures an unknown block using the stored calibration **without** overwriting it. Calibrate on the known reference, then measure others.

3.3 Workflow group

Checkboxes, then fields, then the manual buttons.

Checkbox	Default	What it does
Smart save	off	Detects the injection *while capturing* and writes only those frames. Faster and much smaller, but keeps no full burst
Edge detection V2	off	Alternative edge model. Leave off unless asked
Reference on preview	off	Draws the reference lines on the live preview
Auto pick frames	off	Ignores *Measure from/to* and picks the steadiest 10 frames itself
Ref. lines on output	off	Draws tip line and centre line on saved measure images. The red target rectangle is always drawn
Find piston V2	off	Much faster piston stroke search (~30×). Same result

Field	Default	Notes
Measure from (Nth) / to (Nth)	`10` / `20`	Which frames to measure, counted from the first frame containing a jet
Piston FPS (if no meta)	empty	**Fallback only.** Leave blank — Capture Piston records real timestamps
Barrel dia (mm)	`6.12`	Injector geometry, used for jet speed and pressure
Orifice dia (mm)	`0.155`	As above

`Find Injection` · `Measure` · `Make Video` · `Find Injection Piston` · `Measure Piston` · `Summary`

These run one step at a time on a folder you choose. Normally you do not need them — the Operation buttons chain everything automatically. Use them to re-analyse an old run.

3.4 Operation group

`Stop Preview` · `Capture` · `Start Monitoring` · `Capture Piston` · **`Capture Gel Injection`** · `Open Last Folder`

> Only one of these can own the camera at a time. If a capture is running, the others refuse to start and the status bar says so.

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4. Procedures

4.1 Spray angle

1. Set **Capture time**, **Exposure**, **ROI** and an optional **Run label**.
2. Press **Capture**. It changes to **Stop Capture** while the burst runs.
3. Fire the injector during the burst.
4. The app then does the rest by itself: **find the injection → measure the angle → make a slow-motion video.**
5. Read the result in the **Output** log and press **Open Last Folder**.

What you get — in `data\measure\<run>\`:

- Annotated frames with green edge lines
- `<run>_angles.csv` — one row per measured frame
- Mean angle and a confidence level in the log

4.2 Piston speed

1. Set the ROI — **`320x120` is recommended**, the stroke needs the frame rate.
2. Check **Barrel dia** and **Orifice dia** match the device under test.
3. Press **Capture Piston** (becomes **Stop Piston**), and fire.
4. Chain: **find the stroke → measure it → make a video.**

What you get — in `data\measure\<run>\`:

- `<run>_speed.csv` — per-frame distance and speed, plus a summary block
- `<run>_distance_time.png` and `<run>_speed_time.png`
- In the log: stroke (mm), peak speed, duration, and peak jet speed + pressure

> The video is built from the **measured** frames, so it shows the green line tracking the piston edge. **Watch it.** Nearly every bad measurement in this pipeline was caught by watching the line, not by reading numbers.

4.3 Gel injection

For firing into a gel block.

1. Set ROI and Run label as usual.
2. Press **Capture Gel Injection** (becomes **Stop Gel**), and fire.

3. Chain: **find the injection → make a slow-motion video.** There is **no** angle measurement — inside a gel block there is no free jet to measure.

What you get — the injection images and `<run>_slowmo.mp4` in `data\injection images\<run>\`. Nothing is written to `data\measure\`.

> Folders are named `gel_<timestamp>_<ROI>_<label>` so gel runs are easy to find.
>
> **New feature — check the first few.** The end of a gel injection is detected differently from a jet in air (the gel stays deformed, so the scene never returns to normal). Confirm the video length looks right for the first runs, and report it if the clip is cut short or runs long.

4.4 Monitoring (many injections)

1. Press **Start Monitoring**.
2. Fire as many times as you need. Each detected injection is saved and measured on its own, one at a time.
3. Press **Stop Monitoring** when finished.

> No videos are made in monitoring mode. Drips and mist are detected and discarded automatically — they are logged but not saved.

4.5 Comparing several runs

1. Press **Summary**.
2. Select the measured run folders (multi-select).
3. A `summary_<timestamp>.csv` is written to the top of `data\measure\`.

It reports, for stroke, peak speed, duration, jet speed and pressure: n, mean, standard deviation, **CV %**, min, median, max.

> **CV % is the number that answers "is this injector repeatable?"** — which is usually why the batch was run. Any folder that had to be skipped is listed **with the reason**, so you always know how many runs were actually averaged.

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5. Reading the results

5.1 Where files go

All output sits under the **Output base folder**:

Folder	Contents
`captures\`	The raw burst, one folder per run
`injection images\`	Just the injection, plus the slow-motion video
`measure\`	Annotated frames, CSV reports and plots
`settings\`	Your calibration and nozzle reference
`app.log`	Every message the app has printed

Run folders are named `<timestamp>_<ROI>_<label>`, prefixed `piston_`, `gel_` or `monitor_` by job. Reports are named after their folder, so a file stays identifiable after being copied out.

5.2 Spray angle — `<run>_angles.csv`

Column	Meaning
`angle_deg`	The reported angle for that frame
`quality` / `confidence`	0–100 score and **High / Medium / Low**
`in_rect`	**OK** or **NG** — did the jet stay inside the target rectangle
`coverage`, `rms_*_px`	How well the fitted line matched the visible edge

Frames scoring **Low** are excluded from the trusted mean. A run with many Low frames usually means the jet had already decayed — capture earlier, or use **Auto pick frames**.

5.3 Piston speed — `<run>_speed.csv`

Column	Meaning
`time_ms`	Real elapsed time from the camera's own timestamps
`distance_mm`	Travel from the resting position
`speed_mm_s`, `speed_m_s`	Piston speed
`stable`	`no` = the detector may have jumped to a different edge on that frame

> Frames marked `stable = no` are **reported, never removed**. Peak speed is simply the largest value in the speed column, so the summary and the plotted curve always show the same number.

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6. If something goes wrong

Symptom	Likely cause	What to do
"No injection detected"	The burst missed the shot, or motion was too small	Fire during the burst; increase capture time
Results in pixels, no millimetres	No calibration stored	Do section 2.2 step 1
Angle looks wrong / no `in_rect`	No nozzle reference stored	Do section 2.2 step 2
`in_rect` blank, angle looks like edge-to-edge	Reference was set at a **different ROI** than you captured at	Redo **Set Reference** at the ROI you are using (2.2 step 2)
Millimetres look wrong but nothing failed	Stale calibration — camera, lens or distance changed	Redo **Auto Calibrate** (2.2 step 1)
Angles wrong on every frame	Nozzle is on the **left** of the frame	Re-aim so the nozzle is on the **right**
Green line not on the piston edge	Lighting	The piston must be a **dark shape on a bright barrel**. Adjust the light and re-run
A button does nothing	Another job owns the camera	Read the status bar; stop the running capture first
Frame rate lower than expected	ROI, not exposure	Switch to `320x120`. Raising exposure will not cost you frame rate
Preview stopped by itself	Camera disconnected	Reconnect, then press **Stop Preview** / start again

Before asking for help, collect:

1. `app.log` from the output folder
2. The run folder from `captures\` (or `injection images\`)
3. What you expected versus what you got

7. Quick reference

```
FIRST TIME          Auto Calibrate  →  Set Reference   (in that order)
REDO BOTH IF        camera moved / lens changed / refocused / distance changed
REDO REFERENCE IF   Capture ROI changed, or nozzle refitted
ALWAYS              nozzle on the RIGHT of the frame

SPRAY ANGLE         Capture              → angle + video
PISTON SPEED        Capture Piston       → speed curves + video
GEL INJECTION       Capture Gel Injection→ images + video
REPEATABILITY       Start Monitoring ... Stop Monitoring
COMPARE RUNS        Summary              → CV %

ROI  320x240 = 1048 fps      320x120 = 1798 fps
Exposure does NOT change the frame rate. The ROI does.
```